

Palette components (UI widgets) in Android Studio's Layout:

Layouts (for UI structure)

1. **LinearLayout** – arrange views vertically/horizontally.
 2. **ConstraintLayout** – flexible, modern layout with constraints.
 3. **RelativeLayout** – position views relative to each other.
 4. **FrameLayout** – stack views on top of each other.
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Basic Widgets (core UI components)

5. **TextView** – display text.
 6. **EditText** – input text from user.
 7. **Button** – perform actions on click.
 8. **ImageView** – display images/icons.
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Advanced Input Controls (user selections)

9. **RadioButton** – single option selection.
 10. **RadioGroup** – group of radio buttons.
 11. **CheckBox** – multiple option selection.
 12. **Spinner (Dropdown)** – select from a list.
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Dialogs & Lists (interactive UI)

13. **AlertDialog** – show confirmation/info dialog.
 14. **AutoCompleteTextView** – input with suggestions.
 15. **ListView** – display scrollable list of items.
 16. **GridView** – display items in a grid format.
 17. **Toast** – short popup message.
 18. **Snackbar** – modern message with action button.
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TextView

1. **android:id** – unique ID for referencing.
2. **android:layout_width / android:layout_height** – size.
3. **android:text** – text content.
4. **android:textSize** – size of text.
5. **android:textColor** – color of text.

6. **android:gravity** – text alignment inside the view.
 7. **android:textStyle** – style (bold, italic).
 8. **android:fontFamily** – custom font.
 9. **android:ellipsize** – handling overflow (end, marquee).
 10. **android:lines** / **android:maxLines** – text line limits.
 11. **android:padding** / **android:margin** – spacing.
 12. **android:background** – background color or drawable.
 13. **android:drawableStart** / **drawableEnd** – add icons with text.
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EditText

1. **android:id** – unique ID to reference in code.
 2. **android:layout_width** / **android:layout_height** – size of the input field.
 3. **android:hint** – placeholder text shown when empty.
 4. **android:text** – default text inside the field.
 5. **android:inputType** – type of input (text, number, password, email, phone, etc.).
 6. **android:textColor** – color of the entered text.
 7. **android:textSize** – size of the input text.
 8. **android:ems** – width of the field in terms of character count.
 9. **android:maxLength** – maximum number of characters allowed.
 10. **android:lines** / **android:maxLines** – control number of input lines.
 11. **android:ellipsize** – how text is shown when longer than the box.
 12. **android:drawableStart** / **drawableEnd** – icons inside the input field (e.g., search icon).
 13. **android:padding** / **android:margin** – spacing inside/outside the field.
 14. **android:focusable** / **android:focusableInTouchMode** – controls if field can gain focus.
 15. **android:enabled** – whether user can type or not.
 16. **android:nextFocusForward** – provide previous input id with “@+id/” and that will be next focus component.
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Button

1. **android:id** – unique ID to handle button clicks in code.
2. **android:layout_width** / **android:layout_height** – button size.

3. **android:text** – text displayed on the button.
4. **android:textColor** – color of the button text.
5. **android:textSize** – size of the button text.
6. **android:background** – background color, image, or drawable for button shape.
7. **android:drawableStart / drawableEnd / drawableTop / drawableBottom** – add icons with button text.
8. **android:padding / android:margin** – spacing inside/outside the button.
9. **android:enabled** – enables/disables button interaction.
10. **android:gravity** – alignment of text/icon inside the button.
11. **android:allCaps** – makes text uppercase (default true).
12. **android:onClick** – method in Activity/Fragment that is called when clicked.

```
public void submitForm(View view) {  
    Toast.makeText(this, "Button clicked!", Toast.LENGTH_SHORT).show();  
}
```

ImageView

1. **android:id** – unique ID to reference in code.
2. **android:layout_width / android:layout_height** – size of the image view.
3. **android:src** – sets the image resource (e.g., @drawable/pic).
4. **android:contentDescription** – text description for accessibility (screen readers).
5. **android:scaleType** – defines how the image fits inside (fitCenter, centerCrop, fitXY, etc.).
6. **android:adjustViewBounds** – adjusts the view's bounds to preserve image aspect ratio.
7. **android:background** – background color/drawable behind the image.
8. **android:tint** – apply a color overlay/tint to the image.
9. **android:padding / android:margin** – spacing inside/outside the image view.
10. **android:alpha** – transparency level (0.0 = fully transparent, 1.0 = fully visible).